Comparative Study of Deep Learning Architectures for Atypical Lipomatous Tumor Segmentation in MRI

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1 Introduction

Atypical lipomatous tumors (ALTs) are low-grade soft tissue neoplasms that pose diagnostic and therapeutic challenges in clinical practice. Accurate delineation of tumor boundaries from MRI is essential for treatment planning and follow-up assessment. This report presents a comparative evaluation of four deep learning-based segmentation approaches applied to ALT segmentation from T1- and T2-weighted MRI sequences.

We evaluate two architectural families—the encoder–decoder U-Net with a pretrained backbone (via `segmentation_models_pytorch`, SMP) and the nnU-Net-style PlainConvUNet—each in two input configurations: standard 2D (single slice) and 2.5D (three consecutive slices stacked as channels). All models are trained with 5-fold patient-level cross-validation and assessed on a held-out test set.

2 Methods

2.1 Dataset and Preprocessing

MRI volumes (T1 and T2) were converted to 2D axial slices. Expert-annotated binary masks delineate the tumor region. A patient-level split reserves a held-out test set; the remaining patients are partitioned into 5 folds for cross-validation. All images are resized to 256×256 pixels.

A `slice_tumor_map` flags slices containing tumor. During training, *progressive negative sampling* linearly ramps the fraction of tumor-free (negative) slices from 0 to NEG_SAMPLE_RATIO = 0.5 across epochs, mitigating class imbalance while gradually introducing harder examples.

Data augmentation (training only) includes horizontal/vertical flips, rotations, elastic transforms, brightness/contrast jitter, and coarse dropout via the `Albumentations` library. For 2.5D models, each input is a 3-channel stack of the previous, current, and next axial slice, normalized with ImageNet statistics.

2.2 Model Architectures

Table 1 summarizes the four approaches.

Table 1: Model architecture summary.

ID	Architecture	Encoder / Config	Input Ch.	Params
M1	SMP U-Net 2D	EfficientNet-B4 (ImageNet) + scSE decoder	1	20.3 M
M2	SMP U-Net 2.5D	EfficientNet-B4 (ImageNet) + scSE decoder	3	20.3 M
M3	nnU-Net 2D	PlainConvUNet, 7 stages, deep supervision	1	29.5 M
M4	nnU-Net 2.5D	PlainConvUNet, 7 stages, deep supervision	3	∼29.5 M

SMP U-Net (M1, M2). Built with `segmentation_models_pytorch`. The encoder is an EfficientNet-B4 pretrained on ImageNet. The decoder uses spatial and channel squeeze-and-excitation (scSE) attention. Transfer learning is employed: the encoder is frozen for the first 5 epochs, then unfrozen with a 10× reduced learning rate for fine-tuning.

nnU-Net PlainConvUNet (M3, M4). Uses the `dynamic_network_architectures` implementation of the nnU-Net plain convolutional U-Net. The network has 7 stages with feature maps $[32, 64, 128, 256, 320, 512, 720]$, 2 convolutions per stage, and stride-2 downsampling. Deep supervision is enabled with geometrically decaying weights across resolution levels. Unlike M1/M2, no pretrained weights are used—the model is trained from scratch.

2.3 Training Protocol

All models share the following training configuration:

- **Optimizer:** AdamW, initial learning rate 3×10^{-4}
- **Scheduler:** Warmup cosine annealing (3 warmup epochs, 50 total epochs)
- **Batch size:** 8
- **Loss:** Combined loss $= 0.8 \cdot \mathcal{L}_{\text{Dice}} + 0.15 \cdot \mathcal{L}_{\text{Focal}} + 0.05 \cdot \mathcal{L}_{\text{Boundary}}$
- **Early stopping:** Patience of 10 epochs on validation Dice
- **Gradient clipping:** Max norm 1.0
- **Mixed precision:** Enabled on CUDA
- **Cross-validation:** 5-fold, patient-level splits

2.4 Inference Enhancements

At test time, two optional enhancements are evaluated:

- **Test-Time Augmentation (TTA):** The input is passed through the model in four orientations (original, horizontal flip, vertical flip, both flips). The resulting logits are averaged before applying sigmoid activation.
- **Post-Processing (PP):** Morphological operations and connected-component filtering to remove small spurious predictions.

2.5 Evaluation Metric

The primary metric is the **Dice Similarity Coefficient (DSC)**, defined as:

$$\text{DSC} = \frac{2|P \cap G|}{|P| + |G|}$$

where P is the predicted segmentation and G is the ground truth.

3 Results

3.1 Best Validation Dice per Fold

Table 2 reports the best validation Dice achieved in each fold during training.

Table 2: Best validation Dice per fold and cross-validation summary.

Model	**Fold 1**	**Fold 2**	**Fold 3**	**Fold 4**	**Fold 5**	**Mean**	**Std**
M1: SMP U-Net 2D	0.6625	0.6851	0.7211	**0.8552**	0.7473	0.7342	0.0671
M2: SMP U-Net 2.5D	0.6246	0.6166	0.7354	0.8305	0.7663	0.7147	0.0828
M3: nnU-Net 2D	0.6631	0.6935	0.6249	0.8301	0.5802	0.6784	0.0848
M4: nnU-Net 2.5D	0.4546	0.6772	0.6606	0.7476	0.5525	0.6185	0.1031

Table 3: Best validation loss per fold.

Model	**Fold 1**	**Fold 2**	**Fold 3**	**Fold 4**	**Fold 5**	**Mean**	**Std**
M1: SMP U-Net 2D	0.5469	0.5317	0.3725	**0.4305**	0.3981	0.4559	0.0707
M2: SMP U-Net 2.5D	0.5632	0.5400	0.3343	0.4706	0.3928	0.4602	0.0866
M3: nnU-Net 2D	0.5921	0.5447	0.4041	0.4735	0.4591	0.4947	0.0662
M4: nnU-Net 2.5D	0.6415	0.5824	0.3912	0.5308	0.4621	0.5216	0.0880

Observations. Across all models, Fold 4 consistently achieves the highest validation Dice (0.75–0.86), suggesting that particular patient partition is more favorable. M1 (SMP U-Net 2D) achieves the highest mean validation Dice at 0.7342 ± 0.0671, followed by M2 (SMP U-Net 2.5D) at 0.7147 ± 0.0828. M3 (nnU-Net 2D) follows at 0.6784 ± 0.0848, while M4 (nnU-Net 2.5D) ranks last at 0.6185 ± 0.1031—the lowest mean and highest variance of all models, indicating that combining 2.5D input with training from scratch is particularly challenging.

3.2 Test Set Evaluation

Table 4 reports test Dice for each fold's best model, both without and with TTA + post-processing.

Table 4: Test Dice per fold — without and with TTA + post-processing (TTA+PP).

Model	**Fold 1**	**Fold 2**	**Fold 3**	**Fold 4**	**Fold 5**	**Mean**
Without TTA+PP						
M1: SMP U-Net 2D	**0.8060**	0.7241	0.7781	0.6979	0.6683	0.7349
M2: SMP U-Net 2.5D	0.7274	0.7578	0.6997	0.7123	0.6323	0.7059
M3: nnU-Net 2D	0.7783	0.7650	0.5983	0.7669	0.5803	0.6978
M4: nnU-Net 2.5D	0.6480	0.7501	0.6599	0.6656	0.7032	0.6854
With TTA+PP						
M1: SMP U-Net 2D	0.8212	0.7800	0.8056	**0.8362**	0.7284	**0.7943**
M2: SMP U-Net 2.5D	0.7782	0.7836	0.8049	0.7523	0.7033	0.7645
M3: nnU-Net 2D	0.8066	0.8238	0.6810	0.8233	0.6305	0.7530
M4: nnU-Net 2.5D	0.7793	0.7914	0.7413	0.7923	0.6559	0.7520

3.3 Reproducibility: SMP U-Net 2.5D Second Run

The SMP U-Net 2.5D model (M2) was trained a second time to assess reproducibility. Table 5 compares both runs.

Table 5: SMP U-Net 2.5D (M2): comparison of two independent training runs.

	Validation Dice		**Test Dice (TTA+PP)**	
Fold	**Run 1**	**Run 2**	**Run 1**	**Run 2**
Fold 1	0.6246	0.6404	0.7782	0.7484
Fold 2	0.6166	0.7052	0.7836	0.6998
Fold 3	0.7354	0.7520	0.8049	0.7189
Fold 4	0.8305	0.8569	0.7523	0.5909
Fold 5	0.7663	0.7875	0.7033	0.5145
Mean ± Std	0.7147 ± 0.083	0.7484 ± 0.073	0.7645	0.6545

Run 2 achieves marginally higher validation Dice (0.7484 vs. 0.7147) but substantially worse test Dice (0.6545 vs. 0.7645). This gap suggests potential overfitting in Run 2, where higher validation performance does not generalize to unseen test data. The discrepancy highlights the importance of evaluating on a truly held-out set rather than relying solely on cross-validation metrics.

3.4 Summary Comparison

Table 6 presents the aggregated performance metrics.

Table 6: Overall comparison of all approaches (completed models).

Model	**Val Dice (mean ± std)**	**Test Dice**	**Test Dice (TTA+PP)**
M1: SMP U-Net 2D	0.7342 ± 0.0671	0.7349	**0.7943**
M2: SMP U-Net 2.5D	0.7147 ± 0.0828	0.7059	0.7645
M3: nnU-Net 2D	0.6784 ± 0.0848	0.6978	0.7530
M4: nnU-Net 2.5D	0.6185 ± 0.1031	0.6854	0.7520

4 Discussion

4.1 SMP U-Net vs. nnU-Net

The SMP U-Net models (M1, M2) consistently outperform the nnU-Net models (M3) in both validation and test Dice. This advantage is attributable to:

1. **Pretrained encoder:** EfficientNet-B4 weights from ImageNet provide strong low- and mid-level features, enabling faster convergence and better generalization on small medical datasets.

2. **scSE attention:** The spatial and channel squeeze-and-excitation blocks in the decoder help recalibrate feature maps, improving boundary precision.

3. **Encoder freezing strategy:** Freezing the encoder initially prevents catastrophic forgetting of pretrained features, while gradual unfreezing allows domain-specific fine-tuning.

The nnU-Net models, despite having 45% more parameters (29.5M vs. 20.3M), must learn all features from scratch on a relatively small dataset. Deep supervision helps gradient flow but does not fully compensate for the lack of pretrained features. M4 (nnU-Net 2.5D) in particular exhibits the worst validation Dice (0.6185 ± 0.1031) with the highest variance, though its test performance with TTA+PP (0.7520) nearly matches M3 (0.7530), suggesting that post-processing can partially compensate for weaker learned representations.

4.2 2D vs. 2.5D Input

Counter to expectations, the 2.5D variants do not improve over their 2D counterparts:

- M1 (2D) outperforms M2 (2.5D) by +0.030 in test Dice with TTA+PP (0.7943 vs. 0.7645).
- M3 (2D) and M4 (2.5D) achieve nearly identical test Dice with TTA+PP (0.7530 vs. 0.7520), but M3 has substantially better validation Dice (0.6784 vs. 0.6185), suggesting more stable training.

Possible explanations include:

- The inter-slice spacing may be too large for adjacent slices to provide useful contextual information.
- The 2.5D approach triples the effective input dimensionality without adding proportional spatial coherence, potentially diluting informative features with noise from adjacent slices that may not contain tumor.
- With a small dataset, the 2.5D approach may introduce additional variance that hinders generalization.

4.3 Impact of TTA and Post-Processing

TTA + post-processing consistently improves test Dice across all models:

- M1: +0.059 ($0.7349 \rightarrow 0.7943$)
- M2: +0.059 ($0.7059 \rightarrow 0.7645$)
- M3: +0.055 ($0.6978 \rightarrow 0.7530$)
- M4: +0.067 ($0.6854 \rightarrow 0.7520$)

The improvement ranges from 5.5 to 6.7 percentage points across all models, confirming that TTA+PP is architecture-agnostic and should be adopted as standard practice. Notably, M4 benefits the most from TTA+PP, likely because its weaker base predictions have more room for improvement through augmentation-based ensembling.

4.4 Fold Variance and Generalization

A notable observation is the high variance across folds. Fold 4 consistently achieves the best validation Dice (0.83–0.86) across all models, while Fold 5 tends to perform worst. This likely reflects the heterogeneity of the patient population and the small dataset size, where particular patient combinations can strongly influence fold-level metrics.

The M2 reproducibility experiment further underscores this: despite random seed control via `KFold(random_state=42)`, stochastic training dynamics produce meaningfully different test outcomes between runs.

5 Conclusion

Among the four evaluated approaches, **M1 (SMP U-Net 2D)** with test-time augmentation and post-processing achieves the best overall performance with a mean test Dice of **0.7943**. Key findings:

1. **Pretrained encoders matter:** SMP U-Net with ImageNet-pretrained EfficientNet-B4 outperforms the nnU-Net trained from scratch, despite the latter having more parameters and deep supervision.

2. **2.5D input does not help:** Stacking adjacent slices as channels does not improve segmentation quality in this setting, possibly due to inter-slice spacing or dataset size limitations.

3. **TTA+PP is essential:** Test-time augmentation with post-processing provides a consistent $\sim$6% Dice improvement at no training cost.

4. **Small dataset challenges:** High fold-level variance and the reproducibility gap in M2 highlight the need for larger datasets or more robust regularization strategies.

Future work should investigate (i) 3D volumetric architectures for true spatial context, (ii) self-supervised or foundation model pretraining on medical data, (iii) ensemble strategies combining the best fold models, and (iv) larger or multi-institutional datasets to reduce fold-level variance.